
Amazon ML Challenge 2026

Business Entity Resolution at Scale

Methodology, Candidate Generation, Model Architecture and Results

Team_Sumukh

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Final submission	COMB-v2 : a learned combination of two complete, independent pipelines
Public leaderboard (macro $F_{0.5}$)	0.98476
Held-out macro $F_{0.5}$	0.9914 on the 87,911 training entities held out by both pipelines (precision 0.9981, recall 0.9773)
Candidate pairs per Source 1 record	3.68 on the test set (6.38M pairs), fewer than either pipeline alone (6.2 and 6.5)
Test Source 1 records linked	94.2% in every country, with 3.29–3.39 links each
Models and data	open-weight MIT / Apache-2.0 models of at most 278M parameters, fine-tuned only on the provided training data; no external data and no hosted APIs

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Requirement	Where it is answered
(a) Methodology used	Section 1 (summary) and Section 2 (problem analysis, metric, validation design, system overview, development process)
(b) Candidate generation / blocking strategy	Section 3: pipeline A's sparse search and learned filter, pipeline B's learned dense search, the exploratory multi-key blocking, and the submitted candidate set
(c) Model architecture and feature engineering	Section 4: features, cross-encoders, learners and losses, decision rules, the combination model and leakage control
(d) Other relevant information	Section 5 (results, error analysis, the unseen country), Section 6 (compute, scalability, reproducibility, licences, negative results, limitations) and the appendices

Table 1. Where each documentation requirement is answered.

1 Executive Summary

The task is to link every business record in Source 1 (S1) to the records of the same business in Sources 2 and 3 (S2/S3). The sources share no identifiers; names and addresses are noisy, abbreviated, transliterated or missing; about a quarter of the S2/S3 records match nothing; and the test set adds a third country, France, for which no training links exist. Predictions are scored by macro $F_{0.5}$ over S1 entities, and the organisers also rank smaller candidate sets higher.

The team built two complete, independent pipelines, explored a third, and combined the two strongest with a learned second stage (Figure 1):

- **Pipeline A** (Sumukh Chourasia): an IDF-weighted sparse search split by region and a learned candidate filter (6.5 candidates per S1 record), followed by a two-stage LightGBM matcher whose 83 features include the pair scores of three fine-tuned transformer cross-encoders. Test-like evaluation $F_{0.5}$ 0.9887; public leaderboard 0.978 for its version with two cross-encoders.
- **Pipeline B** (Sativik Barhanpure): a fine-tuned multilingual-e5-small bi-encoder with per-country FAISS search over a neighbour graph, a percentile rule and a logistic-regression graph filter (5.95 candidates per S1 record), followed by an XGBoost matcher led by a fine-tuned cross-encoder. Held-out $F_{0.5}$ 0.9875; public leaderboard 0.980.
- **Pipeline C** (Jatin Patin): an exploratory multi-key blocking and GPU LightGBM baseline that established a validator-checked end-to-end path and the memory-bounded engineering needed at full scale. It was not part of the final submission.
- **The combination (COMB-v2)**: the union of the candidate sets of A and B, minus the pairs that both pipelines reject, leaves 3.68 candidates per S1 record, fewer than either pipeline alone. A LightGBM that reads both pipelines' probabilities and their competition context decides each entity's match list. Trained only on the 87,911 training entities that both pipelines held out, it scores 0.9914 there against 0.9885 (A) and 0.9877 (B). Public leaderboard: **0.98476**.

System	Held-out data	$F_{0.5}$	Cand./S1 (test)	Leaderboard
Pipeline A (M-v29; M-v28 on the leaderboard)	test-like evaluation slice, 441,521 entities	0.9887	6.5	0.978
Pipeline B (R5)	its own 10% hold-out, 220,140 entities	0.9875	5.95	0.980
Pipeline A alone (M-v29)	shared hold-out, 87,911 entities	0.9885	6.5	
Pipeline B alone (R6)	shared hold-out	0.9877	6.18	
Mean of the two probabilities	shared hold-out	0.9905	7.48	
COMB-v1: learned combination	shared hold-out	0.9914	7.48	0.98462
COMB-v2: + pre-filter (submitted)	shared hold-out	0.9914	3.68	0.98476

Table 2. Final results at a glance. Scores on different held-out data are not comparable; the shared hold-out is out-of-sample for both pipelines.

1.1 Key ideas

- **Competition, not independent pairs.** No S2/S3 record belongs to two S1 entities. Every matcher therefore sees how a candidate compares with the rival S1 records that want the same record (ranks and margins), and each record is finally assigned to at most one entity.
- **Validation that resembles the test.** The test set has about twice as many distractor records per S1 entity as the training set. Pipeline A trains and validates in a test-like universe built from the training data; pipeline B uses per-country percentile rules and a label-free calibration for the unseen country.
- **Small candidate sets by learning, not by truncation.** Learned filters cut pipeline A's search output from 48.1 to 6.5 candidates per S1 record and pipeline B's neighbour graph from 20.7 to 5.95, each at a negligible cost; a pre-filter on the union then halves the submitted set to 3.68.

- **Cross-encoders as features.** Transformers that read both records of a pair together gave the largest single gain in both pipelines: B's held-out $F_{0.5}$ rose from 0.9812 to 0.9856, and A's from 0.9854 to 0.9885 (leaderboard 0.970 to 0.978).
- **Diversity, combined by a model.** Sparse token search and dense embedding search find different true links (union recall 99.31% against 98.1–98.2% for each). A small LightGBM trained on entities that neither pipeline had seen turned this into +0.0045 on the leaderboard over the better single pipeline.
- **Out-of-sample scores everywhere.** Every score consumed by a downstream model comes from a model that never trained on that entity (disjoint splits or cross-fitting), so held-out estimates transfer to the test set (Section 4.6).

1.2 Team contributions

Member	Contribution
Sumukh Chourasia	Pipeline A end to end: normalisation with learned aliases, region-split sparse search, learned candidate filter, 83-feature two-stage LightGBM with three cross-encoders and decision rules; the test-like validation universe and the unseen-country study; the combination (COMB-v1 and COMB-v2); experiment log, AWS and Kaggle infrastructure, and the submission package.
Satvik Barhanpure	Pipeline B end to end on Kaggle GPUs: fine-tuned bi-encoder blocking with a hard-negative round, per-country FAISS neighbour graph, percentile rule and graph filter, cross-encoder, XGBoost matcher with competition margins, expected- $F_{0.5}$ selection and label-free France calibration (runs R1–R5; the re-run R6 supplied the scores for the combination); results export and this report.
Jatin Patin	Pipeline C: multi-key blocking and a GPU LightGBM baseline with an $F_{0.5}$ -tuned dual-threshold rule; a scale-safe rewrite (bounded TF-IDF, integer pair indices, chunked scoring) that ran full test inference with validator-checked outputs; experiment log.

Table 3. Contributions of the members who developed pipelines.

2 Methodology

2.1 Problem and data analysis

Before modelling, the team measured the structure of the data. Table 4 summarises the files; Table 5 lists the findings that shaped the design.

	Training	Test
Source 1 records (entities)	2,206,821 (India 883,188; US 1,323,633)	1,732,544 (India 809,986; US 663,106; France 259,452)
Source 2 and 3 records	10,320,219 (India 4,133,346; US 6,186,873)	9,969,589 (India 4,717,565; US 3,817,031; France 1,434,993)
True links	7,638,365: 3.46 per entity, at most 5 from S2 and 6 from S3	unknown; about 3.3–3.5 per entity (estimated from record counts)
Entities without a match	123,247 (5.6%)	unknown
Distractors (S2/S3 records matching nothing)	26% of records; 1.15–1.31 per S1 entity	about 40% of records; 2.2–2.35 per S1 entity (estimated)
Countries	India, US	India, US and France (no training links)

Table 4. The data at a glance.

Finding (training data unless stated)	Design consequence
No S2/S3 record is linked to two S1 entities; entities have 3.46 matches on average and 5.6% have none.	Competition features, exclusive assignment, and a per-entity set decision that may return an empty set.
39.6% of S1 records share their normalised name with another S1 record (“Primary Care Group” appears 253 times). The hard negatives are same-name businesses at a different address; only 5–6% of distractors share an exact name with an S1 record.	Address, house-number and number-gap evidence; name-ambiguity counts.
Noise: abbreviations and legal-form variants, typos (including digit-for-letter, “Summit”), word order, website-style and “formerly / doing business as” names, invented trade names, Devanagari in 7–13% of Indian S2/S3 names and about 13% of addresses, perturbed house numbers, about 3% missing addresses.	Normalisation with learned aliases, fuzzy and soft-alignment features, multilingual cross-encoders on raw text, a name-only search pass.
Linked records always share the country label.	All search is within a country; statistics (IDF, percentiles, frequent tokens) are per country.
The test set has 2.2–2.35 distractors per S1 entity against 1.15–1.31 in training (derived from per-source record counts: in training, S2 and S3 hold 1,340,997 and 1,340,857 distractors).	Test-like validation universe (pipeline A); the leaderboard used to diagnose the shift.
France has no training links. French S1 names are often “city + generic word + legal form” (“Nantes Maison SARL”), the test covers few cities, and French addresses carry no postal codes.	Country-relative features and rules, no learned aliases for France, label-free calibration.
File row order and the numeric part of the IDs are uncorrelated with the ground truth ($ p \leq 0.001$ over 7.6M links).	No leak to exploit or to guard against.

Table 5. Findings from the data analysis and the design decisions they led to.

2.2 Evaluation metric

For an S1 entity with predicted set P and true set T, precision and recall combine into

$$F_{0.5} = 1.25 \cdot \text{precision} \cdot \text{recall} / (0.25 \cdot \text{precision} + \text{recall}),$$

and an entity with no true match scores 1 only if nothing is predicted for it. The final score is the mean over all S1 entities. Three consequences shaped every pipeline. First, the harmonic mean weights precision four times as heavily as recall (0.8 against 0.2), so a doubtful link is usually worth leaving out. Second, the decision is a set per entity, not a label per pair. Third, every singleton rewards an empty prediction, so the decision rule must be able to predict nothing.

2.3 Validation design

Every number in this report comes from training entities held out from the model being evaluated. The three validation sets differ (Table 6), so scores are compared only within a set.

Used by	Held-out entities	Construction	Purpose
Pipeline A	441,521 (20% of training S1), the same in every run	Scored in a test-like universe: every evaluation entity plus 40% of the other training entities; the rest are removed with their true matches while every distractor stays. Result: 5.80 targets and 2.34 distractors per S1 (test: 5.76–5.82 and 2.34–2.35).	Model and decision-rule selection for every A version
Pipeline B	220,140 (10% of training S1, seed 42)	Training S1 split into ENC 40% (883,682; bi-encoder and cross-encoder), FIT 50% (1,102,999; XGBoost) and VAL 10%; no model of B sees VAL.	Rule tuning and selection for every B run
Combination	87,911 (B's VAL entities inside A's test-like universe)	Out-of-sample for both pipelines; the combiner is cross-fitted over them in 5 folds.	Combination model, pre-filter cut-off, decision rule
Leaderboard	public part of the test set	3 uploads per day.	Confirmation and diagnosis only; nothing tuned on it

Table 6. Validation sets.

The test-like universe came from a leaderboard surprise. Pipeline A's first full-data run scored 0.9813 on held-out entities but 0.96 on the public leaderboard. The per-source record counts explained the gap: each test S1 entity faces about twice the training set's distractor density. The next run, trained and validated in the test-like universe, scored 0.9852 there and 0.971 on the leaderboard. Pipeline A also reports a “doubled-distractor” score that counts each link to a distractor twice, the test's density of same-name lookalikes.

2.4 System overview

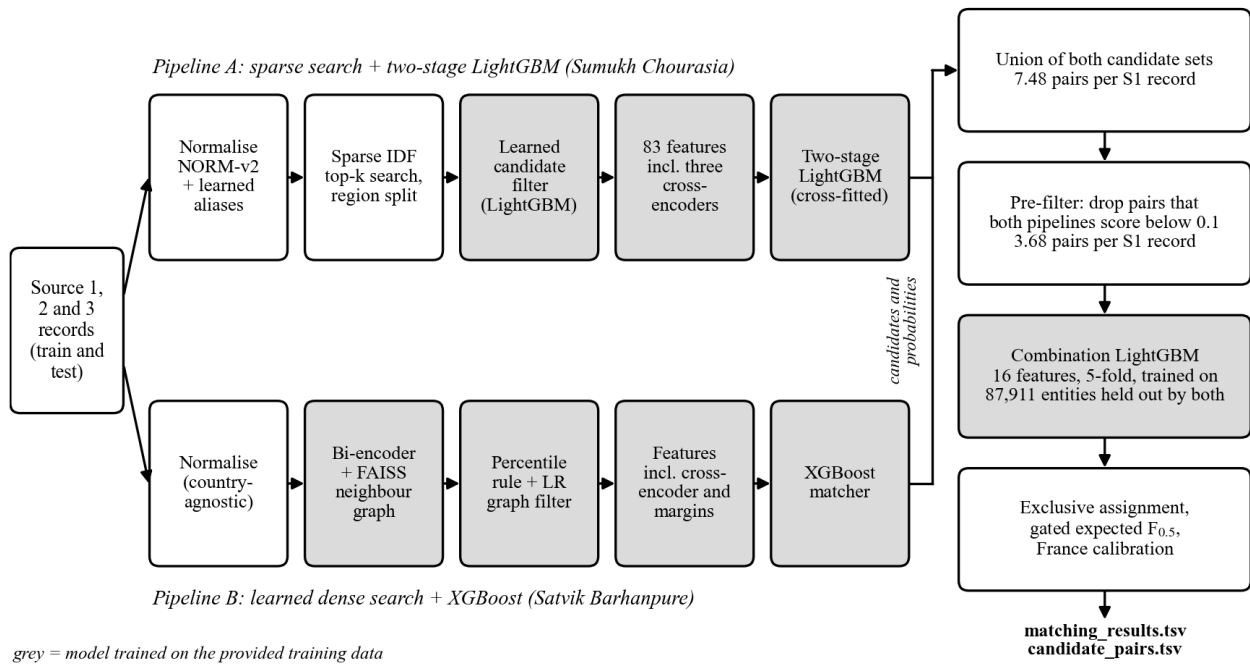


Figure 1. The submitted system. Both pipelines run end to end on their own; the combination reads their candidates and probabilities. Grey boxes are models trained on the provided data.

Both pipelines follow the same pattern (normalise, generate candidates, score pairs with features and gradient-boosted trees, decide per entity) but differ in every component: sparse token search against dense embeddings, LightGBM against XGBoost, three cross-encoders against one. That difference is what makes the combination useful: each pipeline finds true links the other misses. The combination model is trained on the entities that both pipelines held out, so it learns how far to trust each pipeline from out-of-sample scores only.

2.5 Development process

- **Parallel, independent pipelines.** The members developed their pipelines in separate branches and folders of a shared repository (A on the main line, B as Kaggle notebooks, C in *experiments-jatin/*). This kept the pipelines diverse; the combination then exploited that diversity.
- **Versioned components.** Pipeline A names every component version (NORM, BLK, FEAT, MATCH) in *versions.py* and caches each stage under the versions it depends on, so any logged result can be rerun by name. Pipeline B writes every run's arguments and metrics to JSON reports (R1–R6).
- **One change at a time.** Each run changes one thing and is scored on the same held-out entities (for example R1 to R5 in Section 4.3).
- **Experiment logs.** *method_result.md* (A and the combination), *all_approaches.md* (B) and *EXPERIMENTS.md* (C) record every run, including the rejected ones.
- **The leaderboard as a diagnostic.** With three uploads per day, uploads confirmed held-out gains or exposed a shift; no parameter was tuned on the leaderboard.

3 Candidate Generation (Blocking)

Blocking decides which (S1, S2/S3) pairs the matcher sees. Its recall caps the final recall, its size counts in the ranking, and it must scale: the training data alone would give $2.2\text{M} \times 10.3\text{M} \approx 2.3 \times 10^{13}$ pairs. The team built three blocking strategies and submitted a combination of the two learned ones (Section 3.4).

3.1 Pipeline A: IDF-weighted sparse search with a learned filter (Sumukh Chourasia)

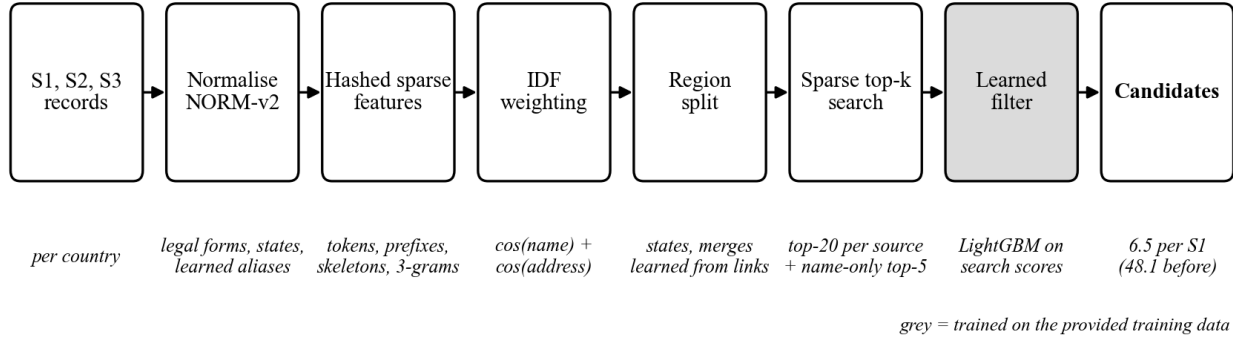


Figure 2. Pipeline A's candidate generation. The grey box is trained on the provided data.

- **Normalisation (NORM-v2).** Transliteration to ASCII (anyascii); canonical legal forms, street types, US states and ordinals; Indian state names joined into one token; French street types; digit-for-letter repairs; “formerly / doing business as” names split; website names turned into words. On top, per-country token aliases learned from the training links (1,105 name and 3,107 address aliases, for example *praivet* → *pvt, mh* → *maharashtra*); France gets none, by design.
- **Blocking keys.** Each record becomes a sparse vector of hashed features: name and address tokens, 4-character prefixes, consonant skeletons, leading and trailing digits of house numbers, the whole joined name, and character 3-grams of the joined name.
- **Scoring and search.** Features are IDF-weighted on the target side; name and address blocks are normalised separately and the score is $\cos(\text{name}) + \cos(\text{address})$. The exact top-k per (S1 record, target source) is found with a multi-threaded sparse top-n product (*sparse_dot_topn*).
- **Region split.** Within a country the search runs inside region groups: frequent tokens that sit at the end of S1 addresses (16 Indian and 37 US states), with merges learned from training links that cross them (Telangana ↔ Andhra Pradesh, Delhi ↔ Haryana). Records without a region token are searched against the whole country; France, having no training links, is searched whole.
- **Name-only pass.** Targets without an address get their own pass, since they would otherwise lose to same-name records that have one.
- **Depth.** Top 20 per S1 record and target source, plus the top 5 targets without an address: 83.3M test pairs (48.1 per S1 record). Recall rises with depth (95.99 / 97.40 / 98.28 / 98.70 / 98.93% at main-pass depths 5 / 10 / 20 / 30 / 40 in the full-train universe), while cutting to the top 5 per source would cost 2.2 points of recall.

Learned candidate filter (BLK-v5-tlu40)

- Most of the 48 candidates are low-ranked rivals: each S1 record has about 3.4 true links. A small LightGBM (63 leaves) sees only what the search produced: the source, the blocking score and its name and address cosines, the name-only flag, and the competition context of those scores (the candidate's rank in its record's list, the gap to the record's best, how many S1 records retrieved the target, this record's rank among them and its margin over the best one). It computes no string similarity, so it stays cheap.

- It is cross-fitted over three folds of the fit entities. Its threshold keeps 99.5% of the true links that the search found; it is set out of fold and never on the evaluation slice. Features and the matcher see only the kept pairs, so *candidate_pairs.tsv* is exactly the set the model scores.
- Result: 11.19M test pairs, **6.5 per S1 record** (median 6, 90th percentile 10, maximum 40; 0.03% of records have none). Evaluation recall goes from 98.69% to 98.21%, the perfect-matcher ceiling from 0.9959 to 0.9945, and the matcher's $F_{0.5}$ from 0.9856 to 0.9854: the filter drops mostly links that the matcher would have missed anyway.

Version	What changed	Pair recall	Cand./S1	Verdict
BLK-v1	IDF-weighted sparse token search (tokens, prefixes, skeletons, house-number digits), top-30	97.66%	60.0	baseline
BLK-v2	+ joined name, name 3-grams, name-only pass for targets without an address	99.31%	69.1	recall up; cost about 100× on full data
BLK-v3	BLK-v2 with a fixed document-frequency cap	97.74–98.49%	69	rejected
BLK-v4	+ search split by region (address tail tokens, learned merges)	99.48%	66.9	noisy region keys
BLK-v4b	region keys limited to state tokens at the end of an address	99.52%	65.9	kept
BLK-v4b@20	top-20 per source on the full data (full-train universe)	98.32%	47.5	full density: −1.1 points
BLK-v4b@20-tlu40	the same in the test-like universe	98.69%	47.8 (test 48.1)	search output
BLK-v5-tlu40	+ learned candidate filter	98.21%	6.0 (test 6.5)	used in M-v11 to M-v29

Table 7. Pipeline A's blocking versions. Rows above the rule are measured on the 10% development sample (DEV-10), rows below on the full-data evaluation slice.

3.2 Pipeline B: learned dense search over a neighbour graph (Sativik Barhanpуре)

Every S2/S3 record belongs to at most one S1 entity, so pipeline B asks the blocking question from the S2/S3 side: *which S1 record is this record closest to?* Each S2/S3 record then keeps about one S1 record, which bounds the candidate set by the size of the S2/S3 pool instead of $S1 \times \text{pool}$.

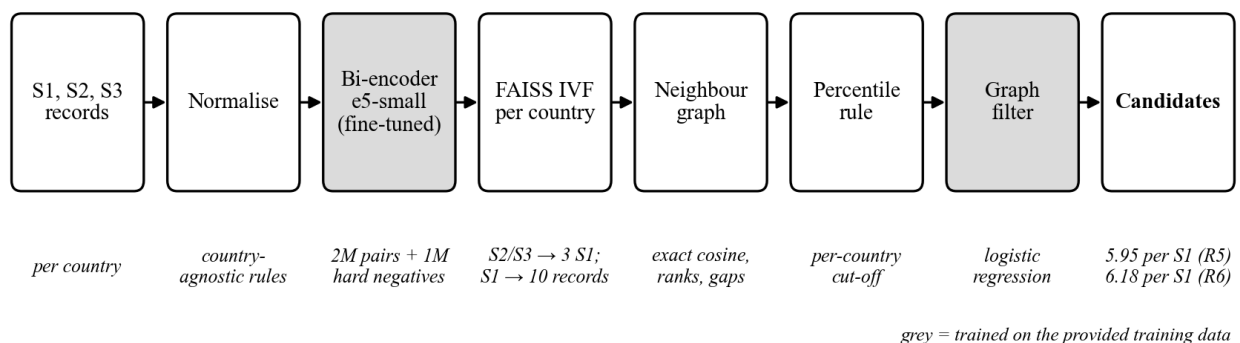


Figure 3. Pipeline B's candidate generation. Grey boxes are trained on the provided data.

- **Bi-encoder.** *intfloat/multilingual-e5-small* (MIT, 118M parameters, 384 dimensions) embeds a record as “query: normalised name ; normalised address”, mean-pooled and L2-normalised. It is fine-tuned for one epoch on 2,000,000 (S1 record, matching S2/S3 record) pairs of the ENC entities with a symmetric in-batch contrastive loss (batch 512, learning rate 0.0001). Half of the batches are drawn from records of one location key (city or state), which makes the in-batch negatives hard: nearby businesses with similar addresses that the model must tell apart by name.

- **Hard-negative round (R5).** A second pass fine-tunes on 1,000,000 mined triplets (S1 record, true match, closest wrong record) at learning rate 3e-05. On validation entities the rule's recall rises from 98.36% to 98.47% while the candidates per S1 fall from 5.39 to 5.31.
- **Approximate search.** FAISS inverted-file indexes with 8-bit scalar quantisation, one per country label (nprobe 64), sharded over two T4 GPUs with a CPU fallback. Each S2/S3 record gets its 3 closest S1 records (reverse neighbours) and each S1 record its 10 closest S2/S3 records (forward neighbours). The union is re-scored with exact cosine similarity: the neighbour graph (18.1 pairs per validation S1, recall 99.03%).
- **Candidate rule.** A graph pair (a, b), with a an S1 record and b an S2/S3 record, becomes a candidate when $level(a, b) \geq \tau$ and $[best(b) - sim(a, b) \leq \delta_{rev} \text{ or } (rank_{fwd}(a, b) < k_{fwd} \text{ and } best(a) - sim(a, b) \leq \delta_{fwd})]$, that is, a is (nearly) the closest S1 record of b, or b is among a's very closest records, and the pair is not among the weakest links of its country. The level is the percentile of the similarity among the closest-S1 similarities of all S2/S3 records of the same country, so the rule adapts to a country whose similarities sit lower overall, such as France. The parameters are chosen on held-out entities as the smallest mean candidate set whose $F_{0.5}$ ceiling (the score of a perfect matcher on the candidates) stays within a tolerance of the best on the grid. R5 uses $\tau = 2\text{nd percentile}$, $\delta_{rev} = 0.05$ and no forward edges.
- **Graph filter.** A logistic regression on seven graph features (percentile similarity; reverse and forward rank; deficit to the best neighbour; gap to the next one) removes about 10% more pairs. Its threshold is the highest one whose $F_{0.5}$ ceiling stays within 0.001 of the unfiltered set (R5: 0.055; weight on the percentile similarity +2.49, all rank and gap weights negative).
- **Result.** R5: 98.16% of held-out true links at 4.78 candidates per validation S1 and 5.95 per test S1 (10,309,277 pairs). R6, the re-run used in the combination (no hard-negative round): 98.09% at 4.89, and 6.18 per test S1 (10,712,138 pairs).

Stage	Recall (VAL)	Cand./S1 (VAL)	$F_{0.5}$ ceiling	Cand./S1 (test)
Neighbour graph (3 reverse + 10 forward neighbours)	99.03%	18.1	0.9973	20.7
Percentile rule, round-1 encoder	98.36%	5.39	0.9952	6.81
Percentile rule, round-2 encoder (hard negatives)	98.47%	5.31	0.9955	6.59
+ graph filter: R5 (stand-alone submission)	98.16%	4.78	0.9946	5.95
R6: round-1 encoder, rule and filter (used in the combination)	98.09%	4.89	0.9944	6.18

Table 8. Pipeline B's blocking stages on its 220,140 validation entities and on the test set.

3.3 Pipeline C: multi-key blocking, exploratory (Jatin Patin)

Pipeline C tested classical key-based blocking with a pairwise gradient-boosted matcher, and the engineering needed to run it on the full test set.

- **Preprocessing.** anyascii transliteration, lower-casing and punctuation removal; “doing business as” prefixes and legal suffixes stripped from names; address stop-words removed; a token map for abbreviations (road → rd); the first digit run as house number and the last 5–6 digit run as postal code.
- **Keys.** Twelve key schemes per country: the first two sorted name tokens, the first token, the full name, 4- and 5-character prefixes, character 4-grams, house number + street, two sorted address tokens, name + house number and name + street. Buckets larger than 300 are pruned and each S1 record keeps at most 80 candidates.
- **Scale engineering.** A TF-IDF fitted on a 300k-record sample with a bounded vocabulary (2^{20} features) and a chunked transform replaced an unbounded one that reached about 40 GB; pairs are held as 64-bit integer positions instead of string columns; test scoring runs in 2M-pair chunks at a flat memory of about 32 GB.

- **Outcome.** The pipeline ran end to end with full test inference (116,968,657 candidate pairs, 67.5 per S1 record) and passed the official validator. Its training runs used a development pool holding about 23% of the S2/S3 records, which capped the reachable recall at 11% of the true links, so its scores measure the pool rather than the method. At 67.5 candidates per S1 record the key-based set is an order of magnitude larger than those of the learned approaches (6.5 and 5.95), so it was not carried into the final submission. Its lesson matches pipeline A's: blocking must be validated at full density.

3.4 The submitted candidate set

The union of both pipelines' test candidates holds 12,964,229 pairs, 7.48 per S1 record (pipeline A 11.19M, pipeline B 10.71M). On the 87,911 shared held-out entities it contains 99.31% of the true links (302,314 of 304,416), against 98.21% and 98.09% for each pipeline alone on its own held-out entities: the sparse and the dense search miss different links.

Most union pairs are low-ranked rivals that neither matcher believes in, yet each counts as a candidate. The final filtering stage therefore drops the pairs that *both* pipelines score below a cut-off before the combination is trained and applied. The selection rule was fixed before the sweep: the largest cut-off whose shared held-out $F_{0.5}$ stays within 0.0001 of the unfiltered combination, stopping where the curve begins to fall.

Cut-off	Held-out pairs / S1	True links kept	Held-out $F_{0.5}$	Test cand. / S1	Test links
none	6.50	302,314 (99.31%)	0.99140	7.48	5,840,814
0.01	3.99	302,250 (99.29%)	0.99138	4.21	5,840,058
0.03	3.81	302,006 (99.21%)	0.99140	3.95	5,838,225
0.05	3.69	301,658 (99.09%)	0.99142	3.81	5,841,028
0.1	3.58	301,115 (98.92%)	0.99139	3.68	5,836,877
0.2	3.51	300,529 (98.72%)	0.99132	3.59	5,838,859
0.3	3.47	299,976 (98.54%)	0.99130	—	—

Table 9. Pre-filter sweep on the shared hold-out (87,911 entities, 304,416 true links). “none” is COMB-v1; 0.1 is COMB-v2.

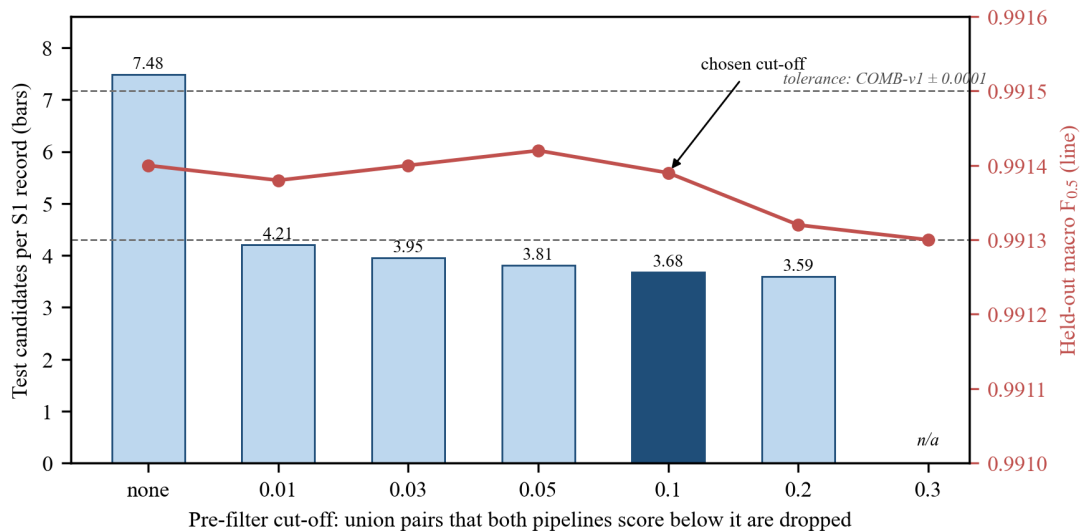


Figure 4. The pre-filter sweep. $F_{0.5}$ stays flat up to 0.1 and falls from 0.2; 0.3 touches the edge of the tolerance band, but the curve is already falling there, so 0.1 was kept.

Result. 6,376,308 test pairs, **3.68 per S1 record**, half of the union and fewer than either pipeline alone (6.5 and 6.2), at the same held-out $F_{0.5}$ (0.99139 against 0.99140). The pre-filter keeps 98.92% of the held-out true links; the ones it drops are links the combination would not have chosen. *candidate_pairs.tsv* is exactly this set, and the combination scores every pair in it.

3.5 Comparison of the blocking strategies

	A: sparse region search	B: learned dense search	C: multi-key blocking	Submitted
Representation	hashed IDF-weighted tokens, prefixes, skeletons, digits and name 3-grams	fine-tuned multilingual-e5-small embeddings (384 dimensions)	12 exact key schemes; TF-IDF features	union of A and B
Search	exact sparse top-k within country and region	FAISS inverted file within country; reverse and forward neighbours	hash buckets, pruned above 300	—
Selection	top-20 per source + name-only top-5, then a learned LightGBM filter	percentile rule, then a logistic-regression graph filter	at most 80 per S1	drop pairs both pipelines score below 0.1
Unseen country	whole-country search, no aliases for France	per-country percentile cut-off	—	—
Held-out recall	98.21% (98.69% before the filter)	98.16% (R5); 98.09% (R6)	not measured at full density	98.92% (99.31% before the pre-filter)
Candidates / S1 (test)	6.5 (48.1 before the filter)	5.95 (R5); 6.18 (R6)	67.5	3.68

Table 10. The three blocking strategies and the submitted candidate set.

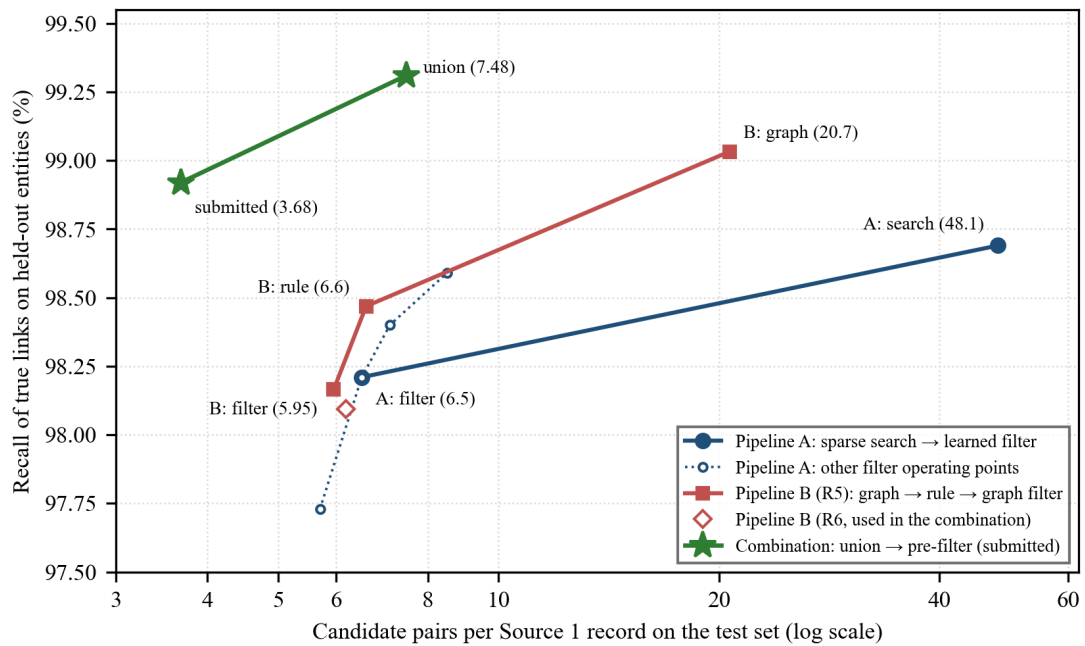


Figure 5. Recall of true links against candidates per test S1 record, stage by stage. Recall is measured on each system's own held-out entities (A: test-like evaluation slice; B: its 10% hold-out; combination: the 87,911 shared entities); every line runs from a stage's input (right) to its output (left).

Scalability. No component compares all pairs. Pipeline A compares each record only with records of its own country and region through a sparse index, so its cost grows with records $\times$ k; pipeline B's inverted-file search is sub-linear in the pool, and each S2/S3 record keeps about one S1 record; the union and the pre-filter are linear in the number of candidate pairs.

4 Model Architecture and Feature Engineering

4.1 Common structure: scoring under competition

All matchers score (S1 record, candidate) pairs with gradient-boosted trees and turn the scores into per-entity match lists in the same steps (Figure 6):

1. **Exclusive assignment.** Because no S2/S3 record belongs to two S1 entities, each candidate record is kept only for the S1 record that scores it highest.
2. **Per-entity set decision.** Either a global threshold, or the expected-F rule: for each entity, the set of its top-ranked candidates (possibly empty) with the highest expected $F_{0.5}$ under the predicted probabilities. The gated variant applies the expected-F rule only when the entity's best probability passes a gate. The rule and its parameter are chosen by macro $F_{0.5}$ on held-out entities.
3. **Unseen-country calibration.** France's probabilities are shifted down in logit space until France leaves the same share of S1 records without a match as the trained countries (Section 5.4).

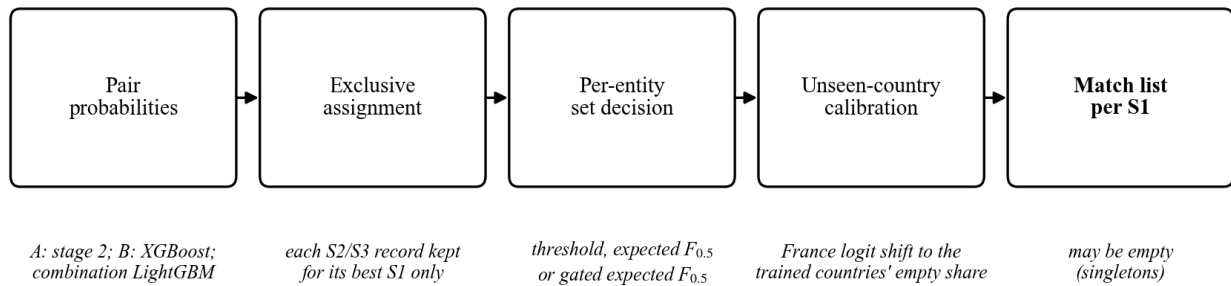


Figure 6. From pair probabilities to match lists, shared by pipeline A, pipeline B and the combination.

4.2 Pipeline A: two-stage LightGBM with three cross-encoders (Sumukh Chourasia)

Stage 1 uses 83 features: 65 string, number and context features and 18 cross-encoder features (Table 11). Stage 2 adds the stage-1 probability and its context.

Family	Features	Purpose
Name similarity	rapidfuzz ratio, token-set, token-sort and partial ratios and Jaro–Winkler on the full, core (legal words removed) and joined names; best match against “formerly / doing business as” parts; token containment both ways	spelling, word order, legal forms
Soft word alignment (FEAT-v2)	each core-name token aligned to its best Jaro–Winkler match in both directions: mean, worst, number of unaligned tokens and the highest IDF among them	a typo against a different business
Address similarity	ratio, token-set, token-sort and partial ratios; containment; house-number Jaccard, share of target numbers present, truncated-number match	same building or only the same street
Number gap (FEAT-v3)	for the closest house numbers found on one side only: digit edit distance, log numeric gap and relative gap	planted neighbours (831 against 835) against digit typos (9052 against 9053)
Distinctive part (FEAT-v4)	name and address similarity after removing the tokens found in more than 1% of the country's own records; tokens left; how many targets share the S1 core name and the target's	generic names, learned without labels, so France is covered
Competition and context	blocking score and cosines; rank within the S1 record (per source and overall); gap to the best; candidate count; S1 records that retrieved the target, this record's rank among them and its margin over the best rival; S1 records sharing the name	at most one S1 per S2/S3 record
Cross-encoders (FEAT-v10)	the probabilities of ce1, ce2 and ce3, each with its rank and margins within the S1 record and against the best rival S1, and the record's top score	a joint, token-by-token reading of both records
Stage 2 only	stage-1 probability, its rank among the record's candidates and among the target's S1 records, the best rival probability and the margins over it	a joint decision across rivals

Table 11. Pipeline A's feature families.

Cross-encoder features

A cross-encoder reads the two records of a pair together (“name | address” of each), so it compares them token by token: typos, transliterations, abbreviations, reordered or swapped words. Each is a one-logit classifier fine-tuned with binary cross-entropy for one epoch on 600k–1.2M pairs. To keep their scores out-of-sample, the fit entities are split into two halves by a hash of the S1 row: the half-A model scores the half-B pairs and vice versa, and every other pair (early stopping, evaluation, test) is scored by a model that never saw its entity.

	Base model	Licence, size	Input text	Trained on	Log loss
ce1 (A)	google/bert_uncased_L-4_H-256_A-4	Apache-2.0, 11M	normalised ASCII	8 CPU cores, cross-fitted halves	0.061
ce2 (A)	intfloat/multilingual-e5-small	MIT, 118M	raw (Devanagari, accents)	Kaggle 2× T4, cross-fitted halves	0.047
ce3 (A)	intfloat/multilingual-e5-base	MIT, 278M	raw	Kaggle 2× T4, cross-fitted halves	0.043
CE (B)	intfloat/multilingual-e5-small	MIT, 118M	name ; address of both records	ENC candidate pairs only (1,500,000 in R5, 2,000,000 in R6)	–

Table 12. The cross-encoders. Log loss is measured alone on pipeline A's evaluation pairs; the full stage-1 LightGBM reaches about 0.040.

Learner, loss and decision rule

- **Model.** LightGBM (MIT), binary log loss, 255 leaves, learning rate 0.1, early stopping on 5% of the entities. Stage 1 and stage 2 are both cross-fitted over three folds of the fit entities (75% of the training entities), so every probability used downstream is out-of-sample. After the filter it trains on 4.2M rows (56.8% positive) in about 10 minutes on 8 cores.
- **Loss.** The decision rule reads the scores as probabilities, so the loss must be a proper scoring rule: binary log loss. Focal loss was not needed (after the filter the classes are balanced and the remaining negatives are hard lookalikes), and exponential loss would let label noise (invented trade names) dominate and gives no

probabilities. The alternatives in Table 13 were tried on the same candidates; none beat plain log loss.

- **Decision.** Exclusive assignment, then gated expected $F_{0.5}$ (gate 0.55 for M-v29).

Variant (same candidates and evaluation slice)	Evaluation $F_{0.5}$
LightGBM, binary log loss (M-v11, kept)	0.9854
distractor rows weighted $\times 2$ (the test's lookalike density)	0.9852
each row weighted by what its error costs its entity's $F_{0.5}$	0.9851
feed-forward neural network (Adam, learning-rate decay, early stopping)	0.9825
average of the network and LightGBM	0.9847
CatBoost	0.9850
average of CatBoost and LightGBM	0.9854
LightGBM + network + CatBoost	0.9851
LightGBM + two cross-encoder features (M-v28)	0.9885

Table 13. Learners and losses tried in pipeline A. Once they saturated, new evidence (the cross-encoders) rather than a new learner moved the score.

4.3 Pipeline B: XGBoost led by a cross-encoder (Satvik Barhanpure)

Family	Features	Purpose
String similarity	rapidfuzz ratio, partial, token-set and token-sort ratios on names, core names and addresses; lengths	surface agreement
IDF token overlap	IDF-weighted cosine of name and address tokens, with IDF fitted per country (France gets its own weights)	rare shared tokens count more
House numbers	agreement of the numbers of both records (token-set, first number, counts)	same building or only the same street
Neighbour graph	bi-encoder cosine and its within-country percentile; reverse and forward ranks, deficits to the best neighbour and gaps to the next	dense-retrieval evidence
Competition margins	each key score minus the best competing pair: the other S1 records claiming the same S2/S3 record (“_bm”) and the other candidates of the same S1 record (“_sm”)	rivalry between S1 records
Name ambiguity	how many S1 and S2/S3 records share the name	generic names are weak evidence
Record context	empty-address flags, candidate counts	record completeness
Cross-encoder	fine-tuned multilingual-e5-small pair probability with its _bm and _sm margins	a joint reading of both records

Table 14. Pipeline B's feature families.

- **Model.** XGBoost (Apache-2.0) on GPU: histogram trees of depth 8, learning rate 0.08, row and column subsampling 0.8, minimum child weight 10, early stopping (692 trees in R6). It is trained on the FIT entities, whose cross-encoder scores come from a model trained only on ENC entities.
- **Selection.** Exclusive assignment, then global thresholds or per-entity expected $F_{0.5}$, whichever wins on VAL (R5: expected $F_{0.5}$, 0.98747 against 0.98733 for thresholds).
- **What the model relies on.** In R6 the cross-encoder features carry 89% of the gain importance and the competition margins 2% (Figure 7). Adding the cross-encoder was the largest single step: 0.9812 to 0.9856 (Table 15).
- **France.** A label-free logit shift makes France's share of empty predictions equal to that of the trained countries (Section 5.4).

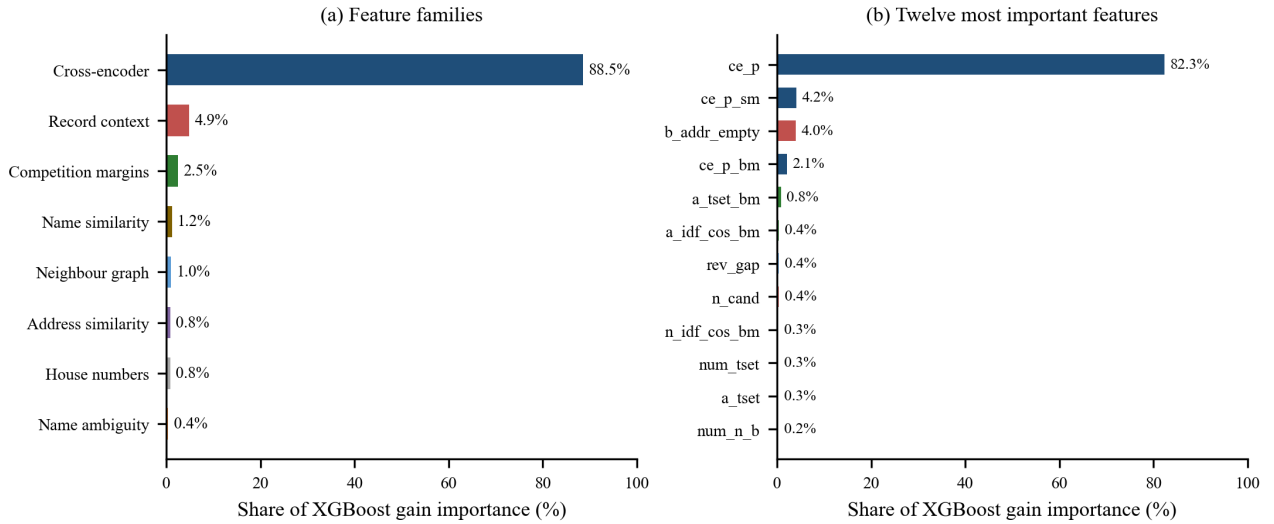


Figure 7. Pipeline B's gain importance in run R6, by feature family (a) and for the twelve most important of its 54 features (b). The cross-encoder family includes the cross-encoder's own margins (ce_p_bm, ce_p_sm).

Run	Change	F _{0.5}	Singletons	India	US	Cand./S1 (test)
R1	XGBoost on string, IDF-overlap and neighbour-graph features	0.9812	0.9788	0.9807	0.9816	6.13
R2	+ cross-encoder score	0.9856	0.9884	0.9860	0.9853	6.13
R3	+ competition margins; empty-share France calibration	0.9867	0.9894	0.9869	0.9866	6.13
R4	+ name-ambiguity counts	0.9871	0.9911	0.9873	0.9870	6.13
R5	+ round-2 encoder (hard negatives); stand-alone leaderboard run (0.980)	0.9875	0.9850	0.9881	0.9870	5.95
R6	re-run without the hard-negative round, longer cross-encoder training; scores kept for the combination	0.9875	0.9935	0.9879	0.9872	6.18

Table 15. Pipeline B, one change per run, on its 220,140 validation entities.

4.4 Pipeline C: GPU LightGBM baseline (Jatin Patin)

- **Features (19).** Six rapidfuzz name and address scores, four token Jaccard and overlap measures, exact-match flags, length ratios, a source flag, and two character TF-IDF (2–4-gram) cosines.
- **Model.** LightGBM 4.7.0 with CUDA (learning rate 0.05, 63 leaves, up to 2,000 trees with early stopping), GroupKFold(5) by S1 entity, negatives subsampled to 20% for fitting while thresholds are tuned on the full out-of-fold predictions. On the GPU, 1M rows $\times$ 100 trees took 2.6 s against 5.0 s on 8 CPU threads.
- **Decision.** A dual threshold, $p \geq t_{\text{high}}$ or $(p \geq t_{\text{low}} \text{ and } p \geq r \cdot p_{\text{max}})$, on a $7 \times 5 \times 4$ grid tuned for out-of-fold macro F_{0.5}, with an optional top-k floor.

Run	Setting	Recall	Out-of-fold F _{0.5}	Runtime	Outcome
EXP-001	baseline; 20k S1 and a 2.4M-record S2/S3 development pool	11.15% of all true links	0.2463	47 min	validator PASS; the pool caps the score (a perfect matcher $\approx$ 0.28)
EXP-002	scale-safe rewrite, same pool	47.86% within the reachable pool	0.1822	53 min	validator PASS; 3.0 links per test entity (training mean 3.46); memory flat
EXP-003	full 10.3M-record pool	—	—	—	no result logged

Table 16. Pipeline C's logged runs (full test inference in each). Development-pool scores are not comparable with the other pipelines.

4.5 The combination model: COMB-v2 (Sumukh Chourasia)

- **Inputs (16 features)** for every pair of the pre-filtered union: each pipeline's probability (missing when its search did not produce the pair) and a presence flag; each probability's competition context (rank and margin over the next candidate of the same S1 record, margin over the best rival S1 record for the same target, the record's top score and candidate count); and the mean and difference of the two probabilities.
- **Training without leakage.** Pipeline B holds out 10% of the training entities from all its models, and pipeline A's scores are out-of-fold or come from models that never saw the entity. The 87,911 entities in both B's hold-out and A's test-like universe (314,853 pre-filtered union pairs) are therefore out-of-sample for both pipelines.
- **Model and rule.** A LightGBM (binary log loss, 63 leaves, early stopping) cross-fitted over these entities in five folds. The rule is chosen on its out-of-fold probabilities: exclusive assignment, then gated expected $F_{0.5}$, gate 0.6. Test pairs get the mean of the five fold models.
- **France.** The same label-free shift as pipeline B's, applied to the combined probabilities (logit -1.39): France then leaves 5.80% of its S1 records empty, as the other countries do, with 3.29 links per record, matching the label-free estimate of 3.31 (3.39 without the shift).
- **Determinism.** Sorted inputs, record-id tie-breaks and deterministic LightGBM make two runs produce byte-identical files, in 2 min 18 s at 9.0 GB peak memory.

Scorer (same 87,911 entities)	Rule	$F_{0.5}$	Precision	Recall	Singletons
Pipeline B alone (R6)	gated expected $F_{0.5}$, gate 0.6	0.98766	0.9973	0.9670	0.9880
Pipeline A alone (M-v29)	threshold 0.75	0.98849	0.9984	0.9680	0.9953
Mean of the two probabilities	threshold 0.65	0.99052	0.9973	0.9769	0.9915
COMB-v1: learned combination	gated expected $F_{0.5}$, gate 0.65	0.99140	0.9982	0.9772	0.9957
COMB-v2: + pre-filter (submitted)	gated expected $F_{0.5}$, gate 0.6	0.99139	0.9981	0.9773	0.9940

Table 17. The combination against each pipeline on the shared hold-out; each rule is chosen on these entities.

Most of the gain is recall at the same precision: each pipeline finds true links the other misses, and when both agree the combination keeps them. The plain mean, which learns nothing, already adds 0.0020 over the better pipeline, so the gain is not a stacking artefact. More inputs for the combiner (28 and 46 features, adding M-v28's probability and the raw cross-encoder scores) changed $F_{0.5}$ by at most +0.00005 and were not adopted: what pipeline A knows is already in M-v29's probability.

4.6 Leakage control

Model	Trained on	Scores consumed downstream
B: bi-encoder and cross-encoder	ENC entities (40%)	FIT, VAL and test pairs, never their own training entities
B: XGBoost	FIT entities (50%)	VAL (10%) for rule selection; test
A: cross-encoders ce1–ce3	two hash halves of the fit entities	each half scored by the other half's model; all other pairs by a model that never saw the entity
A: candidate filter, stage 1, stage 2	fit entities, three-fold cross-fitting	out-of-fold probabilities only
Combination	87,911 entities held out by both pipelines, five-fold cross-fitting	rule chosen on out-of-fold probabilities; test scored by the mean of the fold models
C: LightGBM	GroupKFold(5) by S1 entity	thresholds tuned on out-of-fold predictions

Table 18. Where each model is trained and what it scores. No model's output is ever consumed on the entities it was trained on.

5 Results and Error Analysis

5.1 Scores

Table 19 lists every version with its evaluation data; Figure 8 shows the leaderboard and the held-out progression. The submitted COMB-v2 scores 0.98476 on the public leaderboard, +0.0045 over the better single pipeline (B, 0.980) and +0.0065 over pipeline A's best upload (0.978).

Version	Change	Evaluation data	$F_{0.5}$	Precision	Recall	Leaderboard
M-v1	A: single model, 44 features	DEV-10	0.9882	0.9954	0.9745	
M-v2	+ soft word alignment (52)	DEV-10	0.9894	0.9959	0.9772	
M-v3	+ stage 2, expected-F rule	DEV-10	0.9906	0.9963	0.9794	
dev_v3	+ number-gap features (55)	DEV-10	0.9918	0.9975	0.9810	
FULL-v1	M-v3 on the full data	full-train slice	0.9813	0.9952	0.9544	0.96
M-v5	test-like universe, number gaps, 75% fitted	test-like slice	0.9852	0.9957	0.9644	0.971
M-v6	+ distinctive-part features	test-like slice	0.9856	0.9960	0.9647	
M-v11	+ learned candidate filter (6.5 per S1)	test-like slice	0.9854	0.9962	0.9639	0.970
M-v25	+ cross-encoder ce1	test-like slice	0.9875	0.9979	0.9664	
M-v26	ce2 instead of ce1	test-like slice	0.9884	0.9981	0.9680	
M-v28	ce1 + ce2 with their context	test-like slice	0.9885	0.9983	0.9682	0.978
M-v29	+ ce3 (pipeline A in the combination)	test-like slice	0.9887	0.9984	0.9685	
R1	B: XGBoost on string, IDF-overlap and neighbour-graph features	B's 10% hold-out	0.9812			
R2	B: + cross-encoder score	B's 10% hold-out	0.9856			
R3	B: + competition margins	B's 10% hold-out	0.9867			
R4	B: + name-ambiguity counts	B's 10% hold-out	0.9871			
R5	B: + round-2 encoder (hard negatives)	B's 10% hold-out	0.9875			0.980
R6	B: re-run without the hard-negative round, longer cross-encoder training	B's 10% hold-out	0.9875			
COMB-v1	learned combination of M-v29 and R6	shared hold-out	0.9914	0.9982	0.9772	0.98462
COMB-v2	+ pre-filter, 3.68 per S1 (submitted)	shared hold-out	0.9914	0.9981	0.9773	0.98476

Table 19. Every version with its evaluation data. Rows with different evaluation data are not comparable.

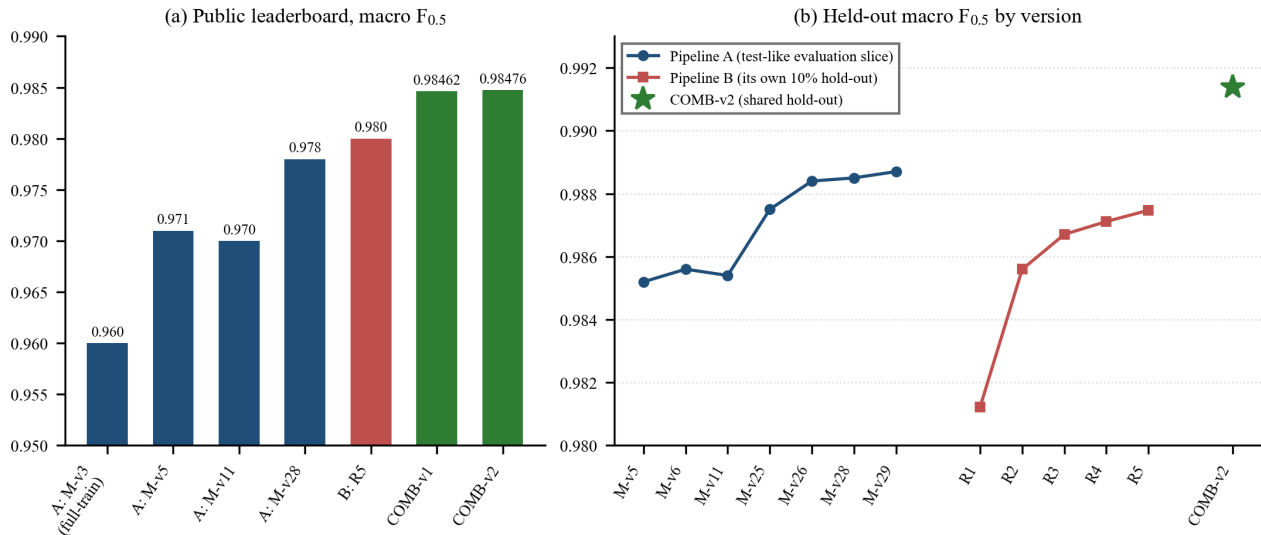


Figure 8. (a) Public leaderboard scores of the uploaded versions. (b) Held-out $F_{0.5}$ by version; the three series use different held-out entities.

5.2 Results by country and segment

System (held-out data)	India	US	Singletons	France on the test set
Pipeline A, M-v29 (test-like slice)	0.9865	0.9902	0.9942	94.1% of S1 linked, 3.20 links each
Pipeline A, M-v28 (test-like slice)	0.9864	0.9900	0.9934	94.2% linked, 3.22 links each
Pipeline B, R5 (own hold-out)	0.9881	0.9870	0.9850	3.17 links each after calibration
Pipeline B, R6 (own hold-out)	0.9879	0.9872	0.9935	3.25 links each after calibration
COMB-v2 (shared hold-out)	—	—	0.9940	94.2% linked, 3.29 links each

Table 20. $F_{0.5}$ by country and for singletons. France has no labels; the last column shows the test-set link profile instead (training truth: 94.4% linked, 3.46 links each).

5.3 Error analysis

Pipeline A's first full-data run (FULL-v1) lost 0.0187 $F_{0.5}$ in total, 0.0145 of it through recall. Table 21 breaks the missed links down; the patterns persisted in later versions, where the cross-encoders recovered part of them.

Error	Share	Typical cases
True link retrieved but not chosen	38% of missed links	replaced or perturbed house numbers; one word swapped (“motors” / “auto”)
True link never retrieved	36% of missed links	initials and acronyms (“chorus vanijya” / “cvprivate”), invented trade names, typo-heavy names
True link taken by another S1 entity	26% of missed links	91% are targets without an address whose name several S1 entities share; mostly irreducible
Wrong link (false positive)	85% lookalike copies	same name at a nearby number; same number with a unit letter (“3027 c douglas ave”); a name variant at the same address; in France, a generic name on another street

Table 21. Error categories of pipeline A (FULL-v1, full-data evaluation slice).

For pipeline B (R5) on its 220,140 validation entities, 13,959 true links were lost in blocking, 11,502 were candidates the matcher did not choose, and 1,961 predicted links were wrong. The matcher trades recall for precision, as $F_{0.5}$ rewards, and the blocking ceiling (0.9946) bounds the score.

5.4 The unseen country: France

France is 15% of the test entities and has no training links, so no held-out score exists for it. Leave-one-country-out runs measured what an unseen country costs: a matcher fitted on the US alone scores India at 0.9444 (0.9817 when India is fitted), and one fitted on India scores the US at 0.9637 (0.9879), a loss of 0.024–0.037 in precision and recall alike. No decision rule recovers it (the best rule on India's own labels: 0.9446). The team therefore built country-relative components and a label-free calibration:

- **Country-relative representations:** per-country IDF weights and percentile cut-offs (B), distinctive-part features learned from each country's own token frequencies (A), no learned aliases for France (A), and multilingual cross-encoders reading the raw text.
- **A label-free false-acceptance diagnostic (B):** when a record is linked confidently to its closest S1 record, its second-closest S1 record is a different business; re-scoring that pair as if the closest were absent reproduces a distractor. Before calibration the matcher accepted 4.8% of such pairs in France against 0.17% in India and 0.07% in the US (R5), which confirmed over-linking in the unseen country.
- **Empty-share calibration:** France's logits are shifted down until its share of S1 records without a match equals the trained countries'. An earlier variant that equalised the diagnostic's acceptance rates over-corrected (France left 6.9–7.7% of records empty against 5.6–5.8%) and was replaced.

System	France links per S1: before → after	Logit shift	France empty share after
Pipeline B, R5	3.40 → 3.17	−2.37	5.72%
Pipeline B, R6	3.52 → 3.25	−2.28	5.84%
COMB-v1	3.39 → 3.31	−1.15	5.82%
COMB-v2 (submitted)	3.39 → 3.29	−1.39	5.80%

Table 22. France calibration. The label-free estimate from the per-source record counts is 3.31 links per French S1 record.

Three further remedies did not move the evaluation: self-training on the unseen country's confident candidates (+0.0014 on India in the leave-one-country-out setting), a hyper-parameter search scored across countries, and keeping the small house numbers that France's frequency filters dropped (19.3% of French candidates looked like the same address with another number, but the final model linked only 0.46% of them). The cross-encoders narrowed the gap: they added 0.003 on the evaluation slice but 0.008 on the leaderboard, consistent with a pretrained multilingual model transferring to an unseen country better than hand-made string features.

5.5 What the leaderboard taught us

1. **0.96 against 0.9813 held out:** the test has twice the distractors per S1 entity. Training and validating in a test-like universe gave 0.971.
2. **The test's distractors are lookalikes of S1 records that are present** (same name, same street, nearby number) as often as the training set's, so the test has twice the hard lookalikes per entity. Copying every distractor to imitate this (M-v7) taught the model to spot the copies and over-linked the test (98.5% of S1 records linked against about 94%); it was not submitted.
3. **The unseen country** is the likeliest remaining gap between evaluation and leaderboard (Section 5.4).
4. **New evidence beat more tuning:** once features, learners and losses had saturated, cross-encoders added 0.008 on the leaderboard (0.970 to 0.978).
5. **Diversity beat tuning a single pipeline:** combining two pipelines that search differently added 0.0045 over the better one (0.980 to 0.98476).

6 Other Relevant Information

6.1 Compute, runtime and scalability

Stage	Hardware	Runtime	Memory
A: full pipeline, preparation to prediction	8 CPU cores (AWS EC2 r7a.2xlarge) or a Kaggle TPU VM (for its RAM)	2–3.5 h	45–60 GB
A: sparse search on the full training data	8 cores	703 s (top-20); 834 s (top-40)	53.2 GB (top-40)
A: matcher after the filter (4.2M rows)	8 cores	about 10 min	–
A: cross-encoders	ce1: 8 CPU cores; ce2 and ce3: Kaggle 2× T4	ce2 about 1.5 h; ce3 about 3 h	–
B: stage 1 from the hard-negative round (R5)	Kaggle 2× NVIDIA T4	142 min	16 GB per GPU
B: stage 2 (features, cross-encoder, XGBoost, selection)	Kaggle 2× T4	93 min (R5); 117 min (R6)	–
B: end to end (R6)	Kaggle 2× T4	4 h 44 min	–
Combination	8 CPU cores	2 min 18 s	9.0 GB
C: development runs with full test inference	CUDA LightGBM	47–53 min	about 32 GB, flat while scoring

Table 23. Hardware, runtime and memory.

- No stage compares all pairs: sparse top-k within region (A), inverted-file search within country (B), hash buckets (C). Candidate generation grows with the number of records, not with their product.
- Test-time candidate sets: 83.3M search pairs reduced to 11.19M (A); 35.8M graph pairs reduced to 10.31M (B, R5); a 12.96M union reduced to 6.38M (submitted).
- Every model is small enough for commodity hardware: gradient-boosted trees, a logistic regression, and transformers of 11M–278M parameters.

6.2 Reproducibility

- Pipeline A's components are named versions; every stage caches its output under the versions it depends on, and presets (M-v1 to M-v29) rerun any logged result by name.
- Pipeline B runs as two Kaggle notebooks (blocking, then matching) generated from the source by a builder script; every run writes a JSON report with its arguments, metrics and timings. Seeds are fixed (42).
- The combination is deterministic: two runs of the final command produce byte-identical files.
- Every submitted file passed the official validator; checksums of all submitted runs are listed in the output manifest. Appendix A lists the commands.

6.3 Models, licences and data policy

Every model is open-weight or trained from scratch, released under MIT or Apache-2.0, and at most 278M parameters. Each is fine-tuned or trained only on the provided training data and runs offline; the pretrained weights are downloaded once from the Hugging Face hub. No hosted LLM API and no external data are used anywhere.

Model or library	Role	Licence	Size
intfloat/multilingual-e5-small	B: bi-encoder and cross-encoder; A: cross-encoder ce2	MIT	118M
intfloat/multilingual-e5-base	A: cross-encoder ce3	MIT	278M
google/bert_uncased_L-4_H-256_A-4	A: cross-encoder ce1	Apache-2.0	11M
LightGBM	A: candidate filter and matcher; the combination; C: matcher	MIT	trees
XGBoost	B: matcher	Apache-2.0	trees
scikit-learn logistic regression	B: graph filter	BSD-3-Clause	7 weights
FAISS, sparse_dot_topn, rapidfuzz, anyascii	search, string similarity, transliteration	MIT, Apache-2.0, MIT, ISC	–
PyTorch, Hugging Face transformers	fine-tuning and inference	BSD-3-Clause, Apache-2.0	–

Table 24. Models and libraries.

6.4 What did not work

Idea	Pipeline	Outcome	Lesson
Validate on a 10% development sample	A, C	overstated blocking recall by 1.1 points (A); pool-limited scores (C)	validate at full density
Validate in the full-train universe	A	0.9813 held out against 0.96 on the leaderboard	match the test's distractor density
Copy every distractor (M-v7)	A	the model learned to spot copies and over-linked the test	rejected; density via weights instead
Fixed document-frequency cap in the search (BLK-v3)	A	recall -0.8 to -1.6 points	rejected
Other losses and learners (M-v12 to M-v18)	A	0.9825–0.9854 against 0.9854	the features were the bottleneck
Self-training, cross-country tuning, French house-number fix (M-v19 to M-v21)	A	no measurable gain	the France gap needs new evidence
Absolute similarity cut-off in the blocking rule	B	not robust to a new country's score scale	percentile rule
Equalising the orphan-acceptance rates for France	B	over-corrected (France 6.9–7.7% empty against 5.6–5.8%)	empty-share matching
More inputs for the combiner (28 and 46 features)	COMB	at most +0.00005	kept 16 features

Table 25. Rejected ideas, kept in the logs so that their lessons are not lost.

6.5 Limitations and future work

- France cannot be scored offline; the gap between held-out scores (0.985–0.991) and the leaderboard (0.970–0.985) is most likely the unseen country together with the test's lookalike density.
- The combiner learns from 87,911 entities (about 4% of the training entities) because the two pipelines were split independently. Aligning their splits would give it several times more training data.
- Next steps: pseudo-labels for France from links on which both pipelines agree confidently; a third, differently built pipeline for more diversity; distilling the cross-encoders to cut inference time.

7 Conclusion

Treating the task as competition for records rather than as independent pair classification, and validating under test conditions, were the decisive choices. Validation had to be corrected twice: a convenient development sample hid a 1.1-point loss of blocking recall that only full-density validation exposed, and full-density validation still missed that the test has twice the distractors per entity, which a test-like universe built from the training data then reproduced.

The organisers' emphasis on small candidate sets added a second lesson: the search needs depth to find hard links, but the matcher does not need to see that depth. Cheap learned filters on the search's own scores cut the candidate sets several-fold at a negligible cost, and a pre-filter on the union of both pipelines halved the submitted set to 3.68 pairs per record.

Once string features, learners and losses had saturated, new kinds of evidence moved the score: cross-encoders that read both records together, trained and scored so that every classifier sees only out-of-sample values. Finally, two good pipelines built differently were worth more together than either tuned further. A small model trained on entities that neither pipeline had seen learned how to weigh them and moved the public leaderboard from 0.980 to **0.98476**, with **3.68** candidate pairs per Source 1 record.

Appendix A Code and reproduction

Path	Contents
code/business_entity_resolution/src/run_pipeline.py	Pipeline A: every stage (prepare, block, features, train, predict); writes the outputs and runs the validator
src/ber/	A's components: normalize, aliases, blocking, features, model, metrics, partition, versions
src/experiments/cross_encoder.py, src/kaggle/run_ce.py	cross-encoders ce1–ce3: export, fine-tuning, cross-fitted scoring
src/experiments/export_scores.py, combine_team.py	A's score export and the combination (COMB-v1, COMB-v2)
src/team_pipeline/	Pipeline B: source (stage1_blocking, stage2_match and their modules), the Kaggle notebook, README and pinned requirements
experiments-jatin/ (branch jatin/aws-s3-ber)	Pipeline C: run_pipeline.py, experiment log, requirements
make_submission.py, output/README.md	validation and packaging of Team_Sumukh_submission.zip; output manifest with checksums

Table 26. Code map of the team repository.

Commands

```
# Pipeline A, preset M-v29 (from code/business_entity_resolution/src/; D = dataset, W = work dir)
python kaggle/run_ce.py # ce2 / ce3 on 2x T4 (CE_CONFIG selects the model)
python run_pipeline.py --data $D --work $W --out $W/out_m29 --preset M-v29 \
  --stages features,train,predict
python -m experiments.export_scores $W NORM-v2__BLK-v5-tlu40__FEAT-v10__MATCH-v6 $W/m29

# Pipeline B (from src/team_pipeline/; Kaggle, 2x T4; R = run dir)
python -m src.stage1_blocking --data-dir $D --out-dir $R/stage1
python -m src.stage2_match --stage1-dir $R/stage1 --data-dir $D --out-dir $R/stage2 \
  --team-name Team_Sumukh --margins --cross-encoder --fit-max-pairs 4000000 \
  --ce-pairs 2000000 --ce-minutes 45

# Combination, COMB-v2 (8 cores, about 2.5 min, 9 GB)
python -m experiments.combine_team --team $R/stage2 --ours-train $W/m29_train.parquet \
  --ours-test $W/m29_test.parquet --truth $D/train/train_ground_truth.tsv \
  --s1 $D/test/test_source1.tsv --out ../../output --threads 8 --france-shift auto \
  --prefilter 0.1 # without --prefilter: COMB-v1

# Pipeline C (repository root, branch jatin/aws-s3-ber)
python experiments-jatin/run_pipeline.py --data-dir $D --output-dir experiments-jatin/output_dev --device gpu
```

Appendix B Hyperparameters

Component	Setting
A: sparse search (BLK-v4b@20)	hashed features, target-side IDF, top-20 per S1 record and target source, name-only top-5, region groups from 16 Indian and 37 US state tokens
A: candidate filter (BLK-v5-tlu40)	LightGBM, 63 leaves, three-fold cross-fitting; threshold keeps 99.5% of the found true links (set out of fold)
A: matcher (MATCH-v6)	LightGBM, binary log loss, 255 leaves, learning rate 0.1, early stopping on 5% of the entities; three-fold cross-fitting of both stages; 75% of the training entities fitted
A: cross-encoders	one-logit heads, binary cross-entropy, one epoch, 600k–1.2M pairs, two cross-fitted halves; ce3 learning rate 3e-5
A: decision (M-v29)	exclusive assignment; gated expected $F_{0.5}$, gate 0.55
B: bi-encoder	multilingual-e5-small, maximum length 64, batch 512, learning rate 0.0001, one epoch, 2,000,000 pairs; hard-negative round: 1,000,000 triplets, batch 256, learning rate 3e-05
B: search	FAISS inverted file with 8-bit scalar quantisation per country, nprobe 64; 3 reverse and 10 forward neighbours
B: rule and filter (R5)	percentile cut-off 2%, δ_{rev} 0.05, no forward edges; logistic-regression filter threshold 0.055 (ceiling tolerance 0.001)
B: XGBoost	histogram trees on GPU, maximum depth 8, learning rate 0.08, subsample 0.8, column sample 0.8, minimum child weight 10, λ 1, 256 bins, seed 42, early stopping
B: cross-encoder	multilingual-e5-small, binary cross-entropy on ENC candidate pairs (1,500,000 pairs and 25 min in R5; 2,000,000 and 45 min in R6)
Combination	LightGBM, binary log loss, 63 leaves, early stopping, five-fold cross-fitting by S1; gated expected $F_{0.5}$, gate 0.6; France shift by empty-share matching; pre-filter 0.1
C: matcher	LightGBM 4.7.0 (CUDA), learning rate 0.05, 63 leaves, up to 2,000 trees with early stopping, GroupKFold(5) by S1, negatives subsampled to 20% for fitting

Table 27. Main hyperparameters.

Appendix C Glossary

Term	Meaning
S1; S2/S3	Source 1 reference records (entities); Source 2 and 3 records to be linked to them
Distractor	an S2/S3 record that matches no S1 entity
Singleton	an S1 entity without a true match; it scores 1 only with an empty prediction
Candidate pair	an (S1, S2/S3) pair produced by blocking, listed in candidate_pairs.tsv
Recall ceiling; $F_{0.5}$ ceiling	share of true links among the candidates; score of a perfect matcher on them
DEV-10	pipeline A's 10% development sample of the training data
Test-like universe (tlu40)	training data rebuilt with the test's distractor density: every evaluation entity plus 40% of the others
ENC, FIT, VAL	pipeline B's split of the training S1 entities: 40% for the bi-encoder and cross-encoder, 50% for XGBoost, 10% for validation
NORM, BLK, FEAT, MATCH; M-vN	pipeline A's versioned components (normalisation, blocking, features, matcher); a preset M-vN combines them
R1–R6	runs of pipeline B (R5: its stand-alone leaderboard run; R6: the re-run whose scores feed the combination)
COMB-v1, COMB-v2	learned combination of A (M-v29) and B (R6), without and with the pre-filter; COMB-v2 is the submission
Cross-fitting	training K models on K – 1 folds each, so every entity is scored by a model that never saw it
Expected-F rule	per entity, the set of top-ranked candidates (possibly empty) with the highest expected $F_{0.5}$ under the predicted probabilities
Orphan acceptance	share of reconstructed distractor-like pairs (a record re-scored with its closest S1 record removed) that the matcher accepts: a label-free false-acceptance rate

Table 28. Glossary.